Numbers with different bases

Decimal (base 10)	Binary (base 2)	Octal (base 8)	Hexadecimal (base 16)
00	0000	00	0
01	0001	01	1
02	0010	02	2
03	0011	03	3
04	0100	04	4
05	0101	05	5
06	0110	06	6
07	0111	07	7
08	1000	10	8
09	1001	11	9
10	1010	12	A
11	1011	13	B
12	1100	14	C
13	1101	15	D
14	1110	16	E
15	1111	17	F

1.3 Number Base Conversions

A binary number can be converted to a decimal by forming the sum of the powers of 2 of those coefficients whose value is 1.

For example, $(1010.011)_2 = 2^3+2^1+2^{-2}+2^{-3} = (10.375)_{10}$

Similarly, a number expressed in base r can be converted to its decimal equivalent by multiplying each coefficient with the corresponding power of r and adding.

The following is an example of octal-to-decimal conversion:

$$(630.4)_8 = 6\times8^2+3\times8^1+0\times8^0+4\times8^{-1} = (408.5)_{10}$$

Example 1.1: Convert decimal 41 to binary. First, 41 is divided by 2 to give an integer quotient of 20 and a remainder of 1/2. The quotient is again divided by 2 to give a new quotient and remainder. The process is continued until the integer quotient becomes 0. The coefficients of the desired binary number are obtained from the remainders as follows:

integer quotient		remainder	coefficient
$41/2 = 20$	+	1/2	$a_0 = 1$
$20/2 = 10$	+	0	$a_1 = 0$